INCH-POUND

MIL-DTL-85537A(AS)
22 September 1997
SUPERSEDING
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16 January 1983

DETAIL SPECIFICATION

TRAILER, MUNITIONS, AERO 51C

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the procurement of an 8500 pound capacity munitions trailer designated as AERO 51C.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Code 414100B120-3, Highway 547, Lakehurst, NJ 08733-5100, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1740

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SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-PRF-10924	-	Grease, Automotive and Artillery
MIL-W-21157	-	Weldment, Steel, Carbon and Low Alloy (Yield Strength 30,000-60,000 psi)
MIL-W-22248	-	Weldment, Aluminum and Aluminum Alloy
MIL-V-62038	-	Vehicles, Wheeled; Preparation for Shipment and Storage Of

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-278	-	Welding and Casting Standard
MIL-STD-410	-	Non-Destructive Testing Personnel Qualification and Certification (Eddy Current, Liquid Penetrant, Magnetic Particle, Radiographic, and Ultrasonic)
MIL-STD-831	-	Test Reports, Preparation Of

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

Naval Air Systems Command (NAVAIRSYSCOM)	Data List 6SE00663, Trailer Munitions

(Copies of specifications, standards and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.3 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-E1417	-	Liquid Penetrant Examination, Standard Practice For (DoD adopted)

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ASTM-E1444 - Particle Examination, Magnetic (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19248-2959.)

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1 - Steel-Structural Welding Code (DoD Adopted)

(Applications for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, Miami, FL 33126.)

2.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.2 and 6.3).

3.1.1 Test report. A test report, delineating the results of first article test (see 4.2), shall be in accordance with MIL-STD-831.

3.2 Design and construction.

3.2.1 Construction. The trailer shall be constructed in accordance with NAVAIR Data List 6SE00663 and all documents listed therein.

3.2.2 Welding. All welding shall be in accordance with MIL-W-22248, Class 3; MIL-STD-278, Class M; or MIL-W-21157, Class 5, as applicable.

3.2.2.1 Welding inspection. During fabrication, all welding shall be inspected in accordance with ASTM-E1417 or ASTM-E1444 by personnel qualified in accordance with MIL-STD-410.

3.2.3 Lubrication. The trailer shall be lubricated with grease in accordance with MIL-PRF-10924.

3.3 Performance requirements.

3.3.1 Functioning. The trailer shall be examined for proper functioning of wheels, brake system and electrical system (see 4.5.2).

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3.3.2 Overload. The trailer, when supported by its running gear, shall withstand a proof load of 17,000 (+100, -0) pounds, applied through the 2 sets of chocks positioned across the open hatchway doors (see 4.5.3).

3.3.3 Brake system.

3.3.3.1 Parking brakes. The parking brakes shall be capable of holding the trailer, loaded with an 8,500 (+100, -0) pound test load, on a 10° grade in either direction (see 4.5.4.1).

3.3.3.2 Service brakes. The service brakes shall be capable of stopping the trailer, loaded with an 8,500 (+100, -0) pound test load, within 30 feet when traveling at 20 miles per hour, on level, dry concrete (see 4.5.4.2).

3.4 Towing force. The towing force required to move the trailer from rest on smooth, dry, level, paved surface, such as brushed concrete, shall not be greater than 50 pounds per ton of maximum gross weight, measured at the drawbar (see 4.5.5).

3.5 Mobility. The trailer shall be subjected to and pass the road test (see 4.5.6.1), the maneuverability test (see 4.5.6.2) and the side slope test (see 4.5.6.3) as specified herein.

3.6 Workmanship. Workmanship shall be in conformance to the design and performance requirements specified herein. Equipment having missing, inoperative, defective, bent, broken or damaged parts, shall be rejected. The installation of hardware parts, such as catches, pins, handles, nuts, bolts, shall avoid damaging the hardware or the mounting surface. Damaged hardware or mounting surfaces shall be treated to provide a continuous protective coating. Screws, nuts, and bolts showing cross threading, mutilation or detrimental or hazardous burrs shall be rejected. Insulated wires shall be formed into cables or ducted. Wires, cables, and brake lines shall be routed and positioned in a protective manner to avoid contact with rough or irregular surfaces and sharp edges. All welds shall be free of harmful defects such as cracks, porosity, undercuts, voids, and gaps. There shall be no evidence of burn through. Fillets shall be uniform and smooth. Angular or thickness misalignment, warpage, or dimensional change due to heat from the welding operation shall be within permitted tolerances. There shall be no damage to adjacent parts resulting from welding.

3.7 Marking. Marking for shipment shall be in accordance with MIL-STD-129.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

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c. Sampling inspection (see 4.4).

4.2 First article inspection. The first article sample shall be performed on one complete trailer when a first article sample is required (see3.1). This examination includes the tests specified in table I under First Article and in the sequence shown.

4.3 Conformance inspection. Each trailer shall be subjected to the tests specified in table I under Conformance, and in the sequence shown.

4.4 Sampling inspection. Trailer components shall be inspected using sampling plans noted on the components' detail drawings or if none are specified, as directed by the procuring activity (see 6.2).

4.4.1 Rejection and retest. When a unit fails the inspections (see 4.2 and 4.3), no item still on hand or later produced shall be accepted until the extent and cause of failure is determined and resolved.

4.5 Methods of inspection.

4.5.1 Examination of product. Each trailer shall be examined visually to determine compliance with the requirements of this specification. Each trailer shall be inspected to determine conformance to the engineering drawings. Particular attention shall be given to routing protection and functioning of the lighting circuits and brake lines. Dimensions and attributes not in compliance with drawing tolerances shall not be accepted.

4.5.2 Functioning. Each trailer shall be examined to determine that wheels are correctly aligned, that all brake systems (including back-up) function and are correctly adjusted, that the electrical system is in working order, that all points requiring lubrication are properly lubricated, and that the wheel bearings are properly adjusted and lubricated. To check alignment, the trailer shall be on a level surface with front wheels straight ahead. Measure the distance between the front wheels at hub height. Mark point where measurement was made. Roll trailer forward until these marks are on the opposite side of the axle (180°). Measure the distance between the wheels at hub height at this point. The difference between the two measurements is the actual toe-in, and shall not be over 0.250 inch. If toe-in is not as stated, adjust to 0 to 0.250 inch. For adjustment, loosen jam nuts on both ends of the tie-rods and rotate tie-rods in or out as required. To adjust wheel bearings, jack or prop trailer to raise wheel off the ground. Spin the wheel and tighten adjusting nut until a heavy drag is felt. Then loosen nut until wheel rotates freely and bearing end play does not exceed 0.002 to 0.003 inch. Install cotter pin through nut and axle and replace grease cap.

4.5.3 Overload test. Place a 17,000 (+100, -0) pound test load across the open hatchway doors for not less than ten minutes. At completion of the test, there shall be no evidence of permanent distortion such as warpage of trailer deck, damage such as dents or breakage of components parts, misalignment of wheels, or other defects detrimental to the intended use.

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TABLE I. First article and conformance inspections.

Sequence	Test	Requirement	Test Paragraph	First Article	Conformance
1	Examination of product	3.2	4.5.1	X	X
2	Functioning	3.3.1	4.5.2	X	X
3	Towing Force	3.4	4.5.5	X	
4	Breakaway	3.4	4.5.5.1	X	
5	Side Slope	3.5	4.5.6.3	X	
6	Service Brake	3.3.3.2	4.5.4.2	X	
7	Mobility	3.5	4.5.6	X	
8	Parking Brake	3.3.3.1	4.5.4.1	X	
9	Overload	3.3.2	4.5.3	X	X

4.5.4 Brake system tests. The trailer shall be loaded with an 8,500 (+100, -0) pound test load consisting of bulk material (steel or concrete) placed across the open hatchway doors. The brake system shall then be tested as specified in 4.5.4.1 through 4.5.4.2.2.

4.5.4.1 Parking brake tests. The trailer shall be loaded with an 8,500 (+100, -0) pound test load and placed on a 10 degree inclined ramp, facing uphill. With parking brake applied, the trailer shall not roll. The loaded trailer shall then be tested facing downhill. This test should be performed after all mobility testing to ensure that the brakes have been burnished sufficiently to achieve a successful test .

4.5.4.2 Service brakes. The trailer shall be loaded with an 8,500 (+100, -0) pound test load and connected to a prime mover. The trailer test brakes shall then be tested as specified in 4.5.4.2.1 and 4.5.4.2.2.

4.5.4.2.1 Brake stopping capability test. Tow the loaded trailer at a speed of 20 mph on a level, dry concrete surface. Stop by applying brakes of towing vehicle thus actuating the trailer

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inertia brake. The trailer shall stop within 30 feet or less of the point where brakes were first applied. The trailer shall not jackknife. At the end of five test cycles, inspect the service brakes and actuating mechanism. Failure to operate satisfactorily or evidence of excessive heating or distortion shall be cause for rejection.

4.5.4.2.2 Back-up test. After passing the brake stopping capability test, the trailer shall be subjected to a back-up force. The service brakes shall hold momentarily, but with continued force, shall release and allow unrestricted rearward travel. The test shall be performed five times.

4.5.5 Towing force test. Place a scale between the prime mover and the towbar of the loaded trailer. The prime mover shall apply a gradual towing force through the scale to the loaded trailer, with its brakes released and with the trailer on level concrete. The trailer shall roll before the scale registers 270 pounds.

4.5.5.1 Breakaway test. Manually tow the loaded trailer and exert a pull on the breakaway cable to actuate the brake system. The wheels should lock, bringing the trailer to a halt.

4.5.6 Mobility tests. During the following tests, the trailer shall be closely observed for its ease of handling. Any undesirable tendency to yaw, sway, dog walk or jackknife shall be cause for rejection.

4.5.6.1 Road tests. The trailer, loaded with an 8,500 (+100, -0) pound test load, shall be subjected to the following road tests:

SURFACE	SPEED	DISTANCE
Level Paved Highway	20 - 25 mph	100 miles
Graded Gravel Road	10 - 12.5 mph	50 miles
Level Course 1/	10 - 12 mph	5 miles

1/ On a level course containing obstructions one inch high, regularly spaced at six foot intervals and at right angles to the direction of travel of the trailer.

4.5.6.2 Maneuverability. Upon completion of road testing (see 4.5.6.1), the loaded trailer shall be towed over level paved highway, at speeds of eight mph through fifty right circle turns and fifty left circle turns, both at maximum cramping angle. It shall then perform twenty-five sudden stops at a speed of 20 mph. At the conclusion of these tests and those described in 4.5.6.1, the vehicle shall be closely examined. Any evidence of excessive wear or damage shall be cause for rejection.

4.5.6.3 Side slope test. At a speed of 5 to 10 mph for a distance of 20 feet, the trailer, loaded with an 8,500 (+100, -0) pound test load, shall traverse at not less than an 8 degree (1.75 inch rise/ft) side slope, without overturning.

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4.6 Magnetic particle inspection test. All steel weldments and castings, when required by the detail drawing, shall be subjected to magnetic particle inspection described in ASTM E1444. Any cracks found are a cause for rejection.

4.7 Penetrant inspection. All aluminum welded joints shall be inspected in accordance with ASTM E1417. Any cracks found are a cause for rejection.

4.8 Non-destructive testing. Magnetic particle and dye penetrant testing shall be performed on all stress weld areas at completion of all first article tests at the discretion of the Government inspector.

4.9 Inspection of packaging. The sampling and inspection of preservation and packing shall be in accordance with MIL-V-62038.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory).

6.1 Intended use. The trailer, with applicable adapters, bolsters, cradles, and fixtures, is intended for use in transporting ordnance from storage to aircraft loading area. This utilization is intended to use to full advantage the capabilities of the AERO 47A loader.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2 and 2.3).

c. Whether first article inspection is required (see 3.1).

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d. Sampling inspection (see 4.4).

e. Selection of applicable levels of preservation and packaging required (see 5.1).

6.3 First article provision. When first article samples are required, the construction of the trailer on the contract should not commence until the samples submitted are pronounced satisfactory by the acquiring activity. The submission of further first article samples on subsequent contracts may be waived at the discretion of the acquiring activity. Approval of first article samples or the waiving of first article tests does not eliminate the requirements of conformance inspection (see 4.3). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations and test approval of the first article. Further information can be obtained from the Commander, Naval Air Warfare Center Aircraft Division, Code 4.8.1.2, M/S 562-3, Highway 547, Lakehurst, NJ 08733-5033.

6.4 Subject term (key word) listing.

Ammunition
Bombs
Loader
Magazine
Ordnance
Vehicle
Weapons

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes..

CONCLUDING MATERIAL

Preparing Activity:
Navy - AS
(Project 1740-0183)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-85537A	2. DOCUMENT DATE (*YYMMDD*) 970922

3. DOCUMENT TITLE
   TRAILER, MUNITIONS AERO 51C

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial: (2) DSN: (*If Applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME COMMANDER NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION	b. TELEPHONE NUMBER (*Include Area Code*) (1) Commercial: (908) 323-2628 (2) DSN: 624-2628
c. ADDRESS (*Include Zip Code*) CODE 414100B120-3 HIGHWAY 547 LAKEHURST, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office, 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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